**Code No: 152AE** **R18**

**JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD**
**B.Tech I Year II Semester (Special) Examinations, January - 2021**
**APPLIED PHYSICS**
**(Common to EEE, CSE, IT)**

**Time: 2hrs** **Max.Marks:75**

**Answer any five questions**
**All questions carry equal marks**
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1. Discuss the de-Broglie's hypothesis. Give in detail the experiment of Davisson and Germer in support of hypothesis. [15]

2.a) Derive time independent Schrödinger's wave equation for a free particle.
b) Show that the energies of a particle in a potential box are quantized. [8+7]

3.a) Derive the relation between the intrinsic carrier concentration and absolute temperature.
b) Explain Hall Effect and its importance. [8+7]

4. Explain how a PN junction is formed and discuss V-I characteristics of PN Junction diode. [15]

5.a) Explain construction and working of LED.
b) What are advantages and disadvantages of LEDs in electronic display? [7+8]

6.a) Explain in detail about Ruby laser.
b) Review the parameters that are commonly used to assess the performance of a detector. [7+8]

7. Explain the purpose of an active medium in a gas laser and describe He-Ne laser. [15]

8. Explain Ferro electricity and Derive Clausius-Mosotti equation. [15]

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